

RouteX India

AI-Powered Digital Freight Marketplace & Intelligent Fleet Management System

MASTER TECHNICAL & ARCHITECTURAL SYSTEM SPECIFICATION REPORT

Document Version: 2.0 (Master Verified Audit)
Target OS / Environment: Linux / Windows / Cloud Kubernetes Environment
Backend Core: NestJS 11 + Prisma ORM 5 + Supabase PostgreSQL
Frontend Web: Next.js 16 (App Router) + React 19 + Tailwind CSS v4
Mobile Applications: React Native & Expo (Android SDK)
AI Inference Engine: Groq Cloud SDK (Llama 3.3 70B Versatile)
GIS Routing & Telemetry: Leaflet, OSRM Engine, Socket.IO WebSockets
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EXECUTIVE SUMMARY & SYSTEM PURPOSE:

RouteX India is an end-to-end digital freight marketplace and intelligent fleet management platform designed for commercial logistics across India. It connects Shippers (Companies and Individuals), Transporters (Fleet Owners), and Drivers into a single real-time ecosystem. RouteX replaces manual broker pricing and phone calls with real-time OSRM GIS routing, Groq Llama 3.3 dynamic freight rate estimation, Socket.IO live GPS telemetry, multi-stop delivery management, electronic Proof of Delivery (ePOD), Razorpay gateway integration, and automated digital wallet escrow settlements.

1. ROUTEX — EXECUTIVE OVERVIEW & BUSINESS ARCHITECTURE

1.1 Industry Problem & RouteX Solution

Commercial road freight in India faces severe fragmentation, unstandardized broker pricing, zero real-time visibility, manual paperwork delays, and payment friction. RouteX solves these bottlenecks by integrating an AI-driven dynamic pricing engine, real-time WebSocket telemetry, automated driver-vehicle dispatch, digital POD verification, and double-entry wallet ledgers.

1.2 User Ecosystem & Primary Roles

- **Shippers (Companies / Individuals):** Book freight, request instant AI rates, track live vehicle telemetry on interactive maps, and receive signed digital PODs.
- **Transporters (Fleet Owners):** Manage vehicle registries, assign drivers to incoming loads, track vehicle maintenance/fuel/tyres, and receive automated wallet settlements.
- **Drivers (Owner-Operators & Employees):** Receive load alerts via mobile app, stream live GPS telemetry, execute sequential multi-stop deliveries, and upload ePOD signatures.
- **Super Admins:** Platform management, national dispatch monitoring, user credential verification, AI telemetry review, and dispute resolution.

1.3 Logistics Scope

RouteX supports both **standard commercial freight** (scheduled or on-demand intra-city and inter-city haulage) and **priority/express logistics** (instant priority matching with 15-minute stationary delay monitoring).

2. REPOSITORY TOPOLOGY & TECH STACK

Application	Directory	Technology Stack	Role & Core Responsibility
routeX-backend	routeX-backend/	NestJS 11, Prisma 5, PostgreSQL, Winston, Socket.IO	Core REST API engine, Groq AI pricing, WebSocket tracking, Razorpay gateway, FCM notifications.
routeX-web	routeX-web/	Next.js 16 (App Router), React 19, Tailwind CSS v4, Leaflet	Multi-role web portal for Shippers, Fleet Owners, and Super Admins.
routeX-driver	routeX-driver/	React Native, Expo, Android SDK	Driver mobile app for load acceptance, live GPS streaming, and ePOD upload.
routeX-mobile	routeX-mobile/	React Native, Expo	Customer mobile application for booking and tracking on mobile devices.
Infrastructure	Root (/), k8s/	Docker Compose, Kubernetes, Nginx	Production container orchestration, SSL proxying, and load balancing.

3. CUSTOM AUTHENTICATION & SECURITY ARCHITECTURE

- RouteX implements a **custom Phone OTP and session rotation engine** (completely purged of Clerk dependencies):
- **Primary Login Method:** Phone Number + 6-Digit OTP SMS dispatch.
 - **Phone Normalization:** Converts Indian mobile formats (9876543210, +91 9876543210) to standard E.164 (+919876543210) using regex `/^\+91[6-9]\d{9}$/`.
 - **Security Controls:** SHA-256 OTP hashing (plain text OTP is never stored in DB), 5-minute strict expiry, 60-second resend cooldown, max 3 verification attempts, and single-use invalidation (verified_at).
 - **Session Rotation:** Issues 7-day JWT access tokens and 30-day SHA-256 refresh tokens saved in the auth_sessions database table. Endpoint POST `/api/auth/refresh` performs secure token rotation.
 - **Multi-Provider SMS Engine (ISmsProvider):** Fast2SMS (Indian OTP gateway), Twilio (Global SMS), MSG91 (Enterprise India), and Console Mock Provider (development mode console logging).

4. GROQ AI PRICING ENGINE & LLM DEEP DIVE

Freight rate estimation is powered by a dual-engine architecture in `routex-backend/src/ai/pricing.service.ts`:

- 1. **Groq LLM Engine:** Queries Groq Cloud SDK using the llama-3.3-70b-versatile model. The service supplies distance, tonnage, truck class, fuel pricing, weather conditions, and traffic level, instructing the LLM to output a strictly formatted JSON object containing estimated price, confidence score, and cost breakdown.
- 2. **Deterministic Local Rule Engine:** If Groq API key is unconfigured or network failure occurs, the service automatically fails over to a local formula engine calculating:

$$\text{Price} = \text{BasePrice} + (\text{Distance_Km} * \text{PerKmRate}) * \text{WeightFactor} * \text{WeatherSurcharge} * \text{TrafficSurcharge} + \text{TollEstimate}$$

5. LIVE GPS TRACKING & WEBSOCKET ENGINE

Event Name	Sender	Target Room	Payload Data Structure	Functional Purpose
driver:join	Driver App	Server	{ driverId }	Registers driver socket mapping for targeted dispatching.
shipper:join	Shipper Web	Server	{ bookingId }	Subscribes shipper socket to live tracking room for a trip.
driver:locationUpdate	Driver App	Shipper, Fleet, Admin	{ driverId, bookingId, latitude, longitude, speed, heading }	Streams real-time GPS telemetry to animate map markers.
booking:delay_detected	Server	Shipper, Admin	{ bookingId, stopOrder, durationMinutes, message }	Emits alert when vehicle stays stationary > 15 mins.
booking:completed	Server	Shipper, Fleet	{ bookingId, status: 'completed' }	Triggers trip completion and releases wallet funds.

6. COMPLETE BUTTON & ACTION INVENTORY

Button Name	Location / Screen	Role	API Endpoint Called	Backend & Database Effect
Send Verification OTP	/login, /register	Public	POST /api/auth/phone/send-otp	Generates 6-digit OTP, stores SHA-256 hash in phone_verifications, sends SMS.
Verify OTP & Access	/login, /register	Public	POST /api/auth/phone/verify-otp	Validates OTP hash, creates user/profile, issues JWT session token.
Calculate Instant Rate	/shipper/book	Shipper	POST /api/ai/pricing	Queries OSRM GIS distance and Groq Llama 3.3 LLM for pricing breakdown.
Confirm & Book Truck	/shipper/book	Shipper	POST /api/bookings	Inserts Booking + booking_stops, emits Socket & FCM alerts.
Accept Load	Driver App	Driver	POST /api/bookings/:id/accept	Assigns driver to booking, updates status to assigned.
Arrived at Pickup	Driver App	Driver	PUT /api/bookings/:id/status	Updates status to at_pickup, broadcasts socket status update.
Start Trip	Driver App	Driver	PUT /api/bookings/:id/status	Updates status to in_transit, initiates live GPS streaming.
Submit ePOD & Complete	Driver App	Driver	POST /api/bookings/:id/epod	Stores signature/photo URL, completes trip, transfers wallet funds.
Add Truck	/fleet/vehicles	Transporter	POST /api/fleet/trucks	Inserts new record into trucks table under company ID.
Assign Driver	/fleet/dispatch	Transporter	POST /api/fleet/assign-driver	Pairs driver_id to trucks.driver_id.
Request Payout	/fleet/wallet	Transporter	POST /api/fleet/wallet/payout	Deducts wallet balance, inserts bank transfer payout transaction.

7. DATABASE SCHEMA & PRISMA ENTITY MODEL

Model / Table	Primary Key	Key Foreign Keys	Purpose & Business Meaning
users & profiles	id (UUID)	id \$→\$ users.id	User identity, name, role (shipper, fleet_owner, driver, super_admin), phone.
phone_verifications	id (UUID)	N/A	Stores SHA-256 hashed 6-digit OTPs, attempts, 5m expiry, 60s resend cooldown.
auth_sessions	id (UUID)	user_id	Tracks active JWT refresh token hashes, device IDs, and revocation status.
Booking	id (UUID)	shipper_id, driver_id, truck_id	Freight order record, status state machine, quoted price, signature URLs.

booking_stops	id (UUID)	booking_id	Multi-stop waypoints, sequence order, lat/lng, per-stop ePOD signature.
trucks & drivers	id (UUID)	company_id, driver_id	Commercial vehicle fleet registry, driver license verification, and availability.
wallets & transactions	id (UUID)	profile_id	Double-entry digital wallet ledger for deposits, escrow holds, earnings, and payouts.
push_tokens	id (UUID)	user_id	FCM device push tokens for multi-platform notifications (android, ios, web).

8. SECOND-PASS VERIFIED CODEBASE TRUTH TABLE

The following truth table presents the code-verified implementation audit across all key RouteX subsystems:

Feature / Capability	Status Classification	Implementation Source File	API / Gateway Route	Technical Audit Notes
Phone OTP Auth	■ FULLY IMPLEMENTED	src/auth/sms.service.ts	POST /api/auth/phone/send-otp	SHA-256 hashed OTPs, 60s cooldown, 5m expiry.
Token Rotation Sessions	■ FULLY IMPLEMENTED	src/auth/auth.service.ts	POST /api/auth/refresh	Stores refresh hashes in auth_sessions table.
Multi-Role RBAC	■ FULLY IMPLEMENTED	src/auth/roles.guard.ts	Guard @Roles()	Enforces shipper, fleet_owner, driver, admin.
AI Freight Pricing	■ FULLY IMPLEMENTED	src/ai/pricing.service.ts	POST /api/ai/pricing	Groq SDK (Llama 3.3 70B) + local formula fallback.
AI Route Optimization	■ MOCK / PLACEHOLDER	src/map/map.service.ts	GET /api/map/route	Uses OSRM GIS engine, NOT an AI LLM model.
AI Fuel Optimization	■ MOCK / PLACEHOLDER	src/fleet/fleet.service.ts	GET /api/fleet/fuel	Manual fuel logs CRUD, no predictive ML model.
AI Driver Matching	■ MOCK / PLACEHOLDER	src/booking/booking.service.ts	POST /api/bookings/:id/accept	Broadcast dispatch + first-accept, no ML ranker.
Priority Logistics	■ FULLY IMPLEMENTED	src/booking/booking.service.ts	POST /api/bookings	Priority FCM alerts, surge rate, 15m delay monitoring.
Multi-Stop Waypoints	■ FULLY IMPLEMENTED	src/booking/booking.service.ts	POST /api/bookings	Sequence order, stop arrival, per-stop ePOD.
ePOD Verification	■ FULLY IMPLEMENTED	src/booking/booking.service.ts	POST /api/bookings/:id/epod	Stores digital signature & photo URLs.
Razorpay Integration	■ FULLY IMPLEMENTED	src/payments/payments.service.ts	POST /api/payments/create-order	HMAC SHA-256 signature verification, refunds.
Digital Wallet Ledger	■ FULLY IMPLEMENTED	src/wallet/wallet.service.ts	GET /api/wallet/balance	Double-entry ledger for deposits, escrow, payouts.
Driver GPS & Socket.IO	■ FULLY IMPLEMENTED	src/tracking/tracking.gateway.ts	Socket driver:locationUpdate	Live room broadcasting & stationary delay alerts.
FCM Notifications	■ FULLY IMPLEMENTED	src/notifications/...	POST /api/auth/push-token	Firestore Admin SDK multi-device messaging.
Fleet Asset Management	■ FULLY IMPLEMENTED	src/fleet/fleet.service.ts	GET /api/fleet/trucks	Vehicle, driver, maintenance, fuel, & tyre CRUD.

End of Master RouteX Technical System Specification PDF Document.